



Nmap Scanning

Use this when: you need live hosts, open ports and service versions on a network you are allowed to test.

Security+ · CySA+ · PenTest+
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AUTHORIZED USE ONLY Scan only networks you own or have **written permission** to test. Practice in your lab or on scanme.nmap.org.

How a SYN scan reads a port

-sS · default when run as root/admin

NMAP	PROBE	TARGET REPLY	STATE	STATE	MEANING
Nmap	SYN →	← SYN/ACK	OPEN	open	An app accepts connections
Nmap	SYN →	← RST	CLOSED	closed	Reachable, nothing listening
Nmap	SYN →	----- no reply	FILTERED	filtered	A firewall blocks the probe
				unfiltered	ACK scan: state unknown
				open filtered	No reply: common with UDP
				closed filtered	Idle scan only

ICMP unreachable also means filtered. On an open port Nmap answers SYN/ACK with RST, so the handshake never completes ("half-open").

Recipes

target = 192.0.2.10

<code>nmap -sn 192.0.2.0/24</code>	Live hosts only
<code>nmap 192.0.2.10</code>	Top 1,000 TCP ports
<code>nmap -p 22,80,443 192.0.2.10</code>	Only these ports
<code>nmap -p- 192.0.2.10</code>	All 65,535 TCP ports
<code>sudo nmap -sS -sV -T4 192.0.2.10</code>	SYN scan + versions
<code>sudo nmap -sU --top-ports 20 192.0.2.10</code>	Top UDP ports (slow)
<code>nmap -Pn 192.0.2.10</code>	Host blocks ping
<code>sudo nmap -A 192.0.2.10</code>	OS + versions + scripts
<code>nmap -sV -oA scan1 192.0.2.10</code>	Save 3 formats

Options by job

DISCOVER	<code>-sn</code> ping only · <code>-Pn</code> skip ping · <code>-n</code> no DNS
TECHNIQUE	<code>-sS</code> SYN · <code>-sT</code> connect · <code>-sU</code> UDP · <code>-sA</code> ACK
PORTS	<code>-p 1-100</code> · <code>-p-</code> all · <code>-F</code> top 100 · <code>--top-ports N</code>
DETECT	<code>-sV</code> versions · <code>-O</code> OS · <code>-sC</code> default scripts · <code>-A</code> all
OUTPUT	<code>-oN</code> · <code>-oX</code> · <code>-oG</code> · <code>-oA</code> all · <code>-v</code> verbose
IPV6	<code>-6</code> 2001:db8::10

NSE scripts: know the risk

safe default discovery intrusive vuln brute
dos exploit

DANGER `vuln`, `brute`, `dos`, `exploit` or `all` can crash services and lock accounts. In scope only.

CAUTION Scripts are not sandboxed: never run unreviewed `.nse` files.

Timing templates `-T0 ... -T5`

T0 paranoid **T1** sneaky **T2** polite **T3** normal **T4** aggressive **T5** insane

T3 is the default. The Nmap book recommends **T4** on fast, reliable networks. **T5** trades accuracy for speed and can overload fragile devices (printers, PLCs, old switches).

Before you scan

- ☐ Written scope: IPs, dates, hours
- ☐ Emergency contact + excluded hosts
- ☐ Start at **T3**, small port range
- ☐ Save output: `-oA`



REMEMBER

- Root/admin → `-sS` by default; normal user → `-sT`
- `filtered` = something dropped the probe
- UDP is slow; expect `open|filtered`

- No `-p` = the **1,000** most common TCP ports
- `-sV` checks the real service, not just the port
- Save every scan with `-oA` for your report